Revisiting the Inclination Distribution of Fast Radio Burst Host Galaxies

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1 Introduction

Fast radio bursts (FRBs) are brief, luminous extragalactic radio transients whose progenitors and emission environments remain uncertain e.g., [1, 2]. Host-galaxy demographics offer a route to identifying their progenitor channels and measuring their volumetric occurrence rates, but the observed host population is filtered by both radio-detection and optical-identification selection effects. [3] presented evidence for one such effect: a deficit of FRBs observed in highly inclined, nearly edge-on host galaxies. If real, this deficit would affect inferred FRB rates, comparisons among host environments, and conclusions about where FRB progenitors reside within their galaxies.

Host inclination provides a physically motivated route to such a bias. An FRB embedded in an inclined disk is viewed through a longer column of its host interstellar medium than an otherwise similar source in a face-on disk. The additional ionized and turbulent material can increase dispersion and multipath scattering. Temporal broadening lowers the peak signal-to-noise ratio of a burst and can move it below the detection threshold of a radio search pipeline [4, 5]. The strength of this effect should depend on the host gas distribution, the position of the source within the galaxy, and the observing frequency and time resolution of the survey.

The initial inclination analysis used 23 localized FRB hosts and found fewer edge-on systems than expected from a matched sample of field galaxies [3]. That result motivated the present expanded analysis, but two limitations require particular attention. First, the localized-host sample available at the time was small. Second, galaxy inclinations are inferred from projected light distributions rather than measured directly. The recovered axial ratio can depend on the point-spread function (PSF), image depth, sky subtraction, deblending, nearby sources, and the adopted surface-brightness model. These effects become especially important for compact or low-surface-brightness hosts.

For randomly oriented axisymmetric disks, $\cos i$ is intrinsically uniform. An observed, flux-limited galaxy sample need not follow that ideal distribution: dust attenuation, surface-brightness selection, finite resolution, morphology cuts, and shape-measurement systematics can each distort it. We therefore test the FRB hosts against an empirical control sample rather than treating a uniform $\cos i$ distribution as the sole observational null. A useful control must resemble the FRB hosts in apparent magnitude and disk-like morphology and must have projected shapes measured with a comparable PSF-aware model.

Here we revisit the inclination distribution with a homogeneous structural analysis of 64 FRB hosts having published localizations. We model predominantly r-band images from the DESI Legacy Imaging Surveys and Pan-STARRS1, together with one Hyper Suprime-Cam field, using empirical PSFs and PSF-convolved Sérsic profiles. After applying the working cuts $m_r \leq 22$ and $q = b/a > 0.2$ and excluding unreliable host-shape measurements, the analysis sample contains 50 hosts. We compare this sample with disk-like galaxies selected from the HSC-PDR2 structural catalog of [6], whose axial ratios were obtained from PSF-convolved single-Sérsic fits. We explicitly assess uncertainties from the profile fit, sky background, and inclination conversion, and use independent image- and fit-level diagnostics to test the robustness of each host measurement. This larger, consistently analyzed sample provides a direct test of whether the inclination-dependent deficit reported by [3] persists under a different structural-analysis pipeline and a new empirical control sample.

2 Methodology

2.1 FRB Host Sample and Imaging Data

We begin with 64 FRB hosts that have a published host localization, paired flux and inverse-variance images, and a successful production fit. Host positions are taken from the published localization literature rather than selected by the output of our image-analysis pipeline. We nevertheless run the Probabilistic Association of Transients to their Hosts (PATH) framework [7] as an independent association cross-check using the reported localization uncertainty and the calibrated sources in each field. We do not impose a uniform pipeline PATH-posterior threshold because several secure literature associations are based on information not contained in the archival image alone. We also do not impose a repeater-status cut; the final working sample contains one known repeater, FRB 20220912A.

For each host we use a $10' \times 10'$ r-band flux image and its inverse-variance map. The 64 production images comprise 51 fields from the DESI Legacy Imaging Surveys [8], 12 from the Pan-STARRS1 3π survey [9], and one deeper HSC-SSP image [10]. The Legacy Surveys are preferred where available because their nominal 5σ point-source depth ($r \simeq 23.9$ AB) is deeper than that of the Pan-STARRS1 stacked survey ($r \simeq 23.2$ AB), although the limiting depth for reliable extended-source morphology is brighter and varies between fields. The HSC image replaces a shallower cutout for one host for which the deeper data provide the more reliable morphology. Pixel scales are measured from the WCS of each image rather than assumed globally.

We require $m_r \leq 22$ to limit the analysis to hosts for which the archival imaging can constrain a two-dimensional light profile. The current tabulation retains 60 of the 64 production hosts under this cut. We then require $q = b/a > 0.2$ because the inclination transformation described in Section 2.5 becomes undefined for $q < q_0$. The joint magnitude and axis-ratio cuts leave 53 candidates. Image- and fit-quality review described in Section 2.4 rejects three additional objects, yielding a 50-host inclination sample. Because the same axis-ratio cut is applied to the field sample, all population statements are conditional on $q > 0.2$.

2.2 Source Detection, PSF Modeling, and Photometry

We run Source Extractor [11] on each flux image with the inverse-variance map as the weight image. A 3σ detection and analysis threshold is used so that faint hosts remain represented in the segmentation map, while `DEBLEND_MINCONT`=0.005 limits the fragmentation of structured galaxies into multiple detections. The resulting catalog and segmentation image define the sources used in all later stages.

PSFEx [12] is then used to construct an empirical PSF from point sources across the $10'$ field. The minimum stellar signal-to-noise threshold is reduced iteratively only when needed to obtain a usable stellar sample; the default stopping criterion is 25 accepted stars. The final PSF model is sampled into a 25×25 pixel stamp and supplied to all subsequent PSF-aware measurements and to `GALFIT`.

A second Source Extractor pass includes the PSFEx model and provides the photometric measurements and shape diagnostics used downstream. We calibrate the field independently against Pan-STARRS1 DR1 and the Legacy Survey DR10 Tractor catalog, using the catalog that provides the larger set of at least three clean matched stars; ties favor Pan-STARRS1. The zero point is the sigma-clipped median of the reference-minus-instrumental magnitudes. When this calibration is unreliable, the host magnitude used for magnitude-dependent analysis is replaced by matched Legacy Survey or Pan-STARRS1 reference photometry.

2.3 Host Selection, Deblending, and Sérsic Fitting

The published host coordinate is the default fitting center. We select the nearest extended Source Extractor detection within $5''$ and abort the fit if no galaxy-like source is present, preventing a noise peak or foreground star from being treated as the host. The independent PATH posterior is retained for quality control but does not override a secure literature host in the production configuration.

The initial fitting region is the host segmentation bounding box padded by 20 pixels. Stars are masked. For every neighboring galaxy whose segmentation island intersects this region, we compare the host–neighbor separation with three times the neighbor's Source Extractor `FLUX_RADIUS`. More distant neighbors are masked; closer neighbors are represented by independent Sérsic components, and the fitting region is expanded until every jointly fitted component has the same boundary padding. This procedure limits both contamination from unmodeled neighbors and unnecessary growth of the fitted sky area. The FRB host is always component 1.

We fit the host with a two-dimensional, PSF-convolved Sérsic profile using `GALFIT` [13, 14]. The centroid, total magnitude, effective semimajor axis R_e, Sérsic index n, projected axis ratio q, and position angle are free. Initial values are taken from the PSF-aware Source Extractor catalog. We constrain $0.5 \leq n \leq 6$, $1.5 \leq R_e \leq 100$ pixels, and $8 \leq m \leq 40$; the latter broad interval is a numerical guard rather than a sample-selection cut. Nearby galaxies selected for simultaneous fitting have their own Sérsic components but do not contribute to the reported host parameters.

The pixel uncertainty image is formed from the inverse variance. Invalid or nonpositive inverse-variance pixels are masked, and the absolute uncertainty scale is compared with the robust sky scatter measured from unmasked pixels. If their ratio lies outside 0.5–2, the uncertainty image is rescaled by a single multiplicative factor while retaining its spatial structure. A constant sky pedestal is fitted simultaneously with the galaxy components; sky gradients are held at zero because the comparatively shallow cutouts do not constrain them robustly. Fits whose sky level departs strongly from the independent reference are repeated with a bounded sky term and are rejected if the sky check still fails.

We also run Statmorph [15] to measure concentration, asymmetry, smoothness, Gini, and M_{20}. These quantities are diagnostics of non-Sérsic structure and are not used to replace the `GALFIT` axis ratio.

2.4 Fit Validation and Adopted Axis Ratios

All 64 production fits are subjected to the same validation suite before the science cuts are applied. The checks measure the local reduced χ^2, residual-flux fraction, Fourier structure of the residual, sensitivity to sky perturbations, radial isophotal geometry, leakage of PSF ellipticity into host shape, magnitude-dependent bias, agreement with an independent AstroPhot Sérsic fit, and visual agreement among the data, model, and residual. Deeper archival imaging is inspected where available. These tests are designed to falsify a host-axis-ratio measurement; no single threshold, including reduced χ^2, automatically admits an object to the paper sample.

For hosts whose free-n, free-sky production fit is unstable, we inspect controlled alternatives in which n and/or the sky is fixed, or a contaminating point source is modeled explicitly. The adopted "winning leg" is recorded before population-level comparisons. Of the final 50 hosts, 37 use the production fit and 13 use a documented refit. Three objects that pass the joint m_r and q cuts are excluded because the fitted object or its geometry is not trustworthy: FRB 20190711A, FRB 20221101B, and FRB 20230930A. Full diagnostic definitions and per-object decisions are provided in the Appendix.

2.5 Inclination Angles and Uncertainties

We convert the adopted projected axis ratio $q = b/a$ to inclination using the oblate-disk relation [16]

$$\cos^2 i = \frac{q^2 - q_0^2}{1 - q_0^2}, \tag{1}$$

where $i = 0°$ is face-on and q_0 is the intrinsic short-to-long axis ratio of an edge-on disk. We adopt $q_0 = 0.2$, following common practice for late-type disks and [3]. Objects with $q \leq q_0$ are excluded rather than assigned a formally saturated $i = 90°$. The sensitivity of the population result to plausible choices of q_0 is tested in the Appendix.

The formal covariance uncertainty reported by `GALFIT` captures statistical noise near the best-fitting model but does not include background or model-choice systematics [14, 17]. We therefore do not multiply the formal axis-ratio uncertainty by an arbitrary constant. Instead, we combine three explicitly defined terms. First, $\sigma_{q,\mathrm{fit}}$ is the formal uncertainty of the adopted fit. Second, we estimate the large-scale background uncertainty as

$$\sigma_{\mathrm{sky}} = 1.4826\,\mathrm{MAD}(\texttt{BACKGROUND}), \tag{2}$$

using local Source Extractor background estimates for clean sources across the field. We re-fit each host with the sky fixed at the best-fitting value plus and minus σ_{sky}, obtaining q_+ and q_-, and define

$$\sigma_{q,\mathrm{sky}} = \frac{|q_+ - q_-|}{2}. \tag{3}$$

This follows the general strategy of estimating background systematics through sky-perturbed re-fits rather than through the local covariance matrix [18].

Third, we include a 5° inclination-model term to represent finite disk thickness, non-axisymmetry, dust, and method dependence not captured by a single Sérsic fit. To avoid applying this term twice, we map it into axis ratio before combining uncertainties. With

$$q(i) = \left[q_0^2 + (1 - q_0^2)\cos^2 i\right]^{1/2}, \tag{4}$$

we define

$$\sigma_{q,5°} = \frac{1}{2}\left|q(i + 5°) - q(i - 5°)\right| \tag{5}$$

and adopt

$$\sigma_q^2 = \sigma_{q,\mathrm{fit}}^2 + \sigma_{q,\mathrm{sky}}^2 + \sigma_{q,5°}^2. \tag{6}$$

For uncertainty propagation we draw q from a Gaussian with width σ_q, truncate draws to $q_0 < q \leq 1$, and transform each draw with Equation 1. The same draws are used to construct the uncertainty envelope of the FRB-host cumulative distribution. This protocol supersedes the preliminary prescription that scaled formal `GALFIT` errors by a factor of 10.

2.6 Matched Field-Galaxy Comparison

We construct the field-galaxy control from the HSC-PDR2 structural catalog of [6]. That catalog fits HSC i-band images with a PSF-convolved single-Sérsic model using Lenstronomy. It therefore does not use the same software as our `GALFIT` analysis, but it fits the same parametric profile family and reports the corresponding projected axis ratio. This is closer methodological parity than comparing our model-derived host shapes with heterogeneous isophotal or catalog moments.

From a reproducible random subset of the public HSC catalogs, we retain fits with `goodfits_flag`=1 and $0.4 < n < 1.5$ to select exponential-disk analogues. We then apply the same photometric and geometric cuts used for the hosts, $r \leq 22$ and $q > 0.2$, leaving 24,450 control galaxies. The HSC structural fit is measured in the i band whereas the matching magnitude is the catalog r magnitude; possible bandpass and resolution effects are examined in the Appendix.

Each HSC axis ratio is converted to $\cos i$ with Equation 1 and the same $q_0 = 0.2$. We draw 10,000 samples of 50 HSC galaxies without replacement and use their cumulative distributions to define the median field expectation and its 16th–84th percentile sampling envelope. We compare the 50 host point estimates with the complete HSC pool using two-sample Kolmogorov–Smirnov

and Anderson–Darling statistics. Because the HSC pool is much larger than the host sample, we also calculate sample-size-matched empirical p-values: for each statistic, the p-value is the fraction of 50-object HSC draws whose discrepancy from the parent HSC pool is at least as large as that measured for the FRB hosts. The hypothesis-test p-values use point estimates; the uncertainty-aware CDF envelopes are shown separately. A uniform $\cos i$ distribution is retained as a visual reference, not used as the observational null, because the matched magnitude, morphological, and q cuts reshape the measured field distribution.

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